

Kannika Armstrong (Jaikham)

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[LinkedIn: kannika-armstrong](#) | [GitHub: A-Kannika](#) | [Portfolio: KannikaA](#)

SKILLS

Programming Languages: Java, Python, JavaScript, SQL, PostgreSQL, MySQL, HTML, CSS, PHP, R, Rust

Frameworks & Libraries: scikit-learn, Pandas, NumPy, Matplotlib, TensorFlow, Streamlit, ReportLab, Plotly, Ollama

Tools: MS SQL Server, Bootstrap, WordPress, TiDB, Git, GitHub, Google Colab, Jupyter Notebook, GitHub Actions, Tableau, Excel, VS Code, Linux, Mac OSX, Windows.

EDUCATION

M.S. in Computer Science & Systems | University of Washington, Tacoma, WA | *Expected June 2026*

- Relevant coursework: Advanced Algorithms, Machine Learning, Applied Distributed Computing, Cryptography, Database Systems Internals

B.S. in Computer Science and Systems – University of Washington Tacoma

B.S. in Physics (Minor in Mathematics) – Chiang Mai University, Thailand

PROJECTS, RESEARCH & OPEN SOURCES

Portfolio website: <https://a-kannika.github.io/KannikaA/> (for additional information and projects)

Creator/Developer | Vizolytic (Sales Analytics & Reporting Dashboard): [GitHub](#)

“A Python and Streamlit-based web application for interactive sales analytics and automated PDF reporting.”

- Architected and deployed a web application from scratch using Streamlit and Python to provide interactive sales analytics. (Live at: [vizolytic.streamlit.app](#))
- Built interactive Plotly visualizations, including U.S. choropleth maps and hierarchical treemaps.
- Engineered data filtering, time series analysis, and CSV export features using Pandas.
- Developed an automated PDF reporting feature (ReportLab) to summarize key business insights.

Retrieval Augmented Generation (RAG) with TiDB Vector Search: [GitHub](#)

“Retrieval-Augmented Generation system for cybersecurity knowledge using vector embeddings and local LLM.”

- Developed RAG pipeline leveraging TiDB's native vector search with cosine similarity for semantic document retrieval.
- Implemented document embedding using BAAI/bge-m3 transformer (1024-dim) and integrated Llama 3.2 via Ollama for context-aware question answering.
- Built a complete data pipeline: embedding generation → TiDB storage → vector similarity search → LLM response synthesis.

Master's Thesis | Optimizing Performance and Efficiency of Lattice-Based Post-Quantum Cryptography Algorithms in Resource-Constrained UAV Environments (In Progress) *University of Washington Tacoma*

“Developing, optimizing, and verifying a secure, quantum-resistant protocol for UAVs.”

- Architecting a secure UAV protocol by integrating Kyber (PQC) and Ascon-MAC.
- Implementing NTT, vectorization, and parallelization to optimize for resource-constrained hardware.
- Simulating protocol performance in Python to benchmark latency, memory, and computation time.
- Conducting formal security verification with ProVerif to ensure quantum resistance and protocol robustness.

PROFESSIONAL EXPERIENCE

Teaching Assistant & Academic Mentor | University of Washington & Pierce College *Sep 2024 – Jun 2025 (UW); 2020 – 2024 (Pierce)*

- Led lab sessions, graded Java programming projects, and mentored students in debugging.
- Mentored 50+ students in Algorithms, OOP (Java/Python), and SQL queries.
- Taught collaborative project workflows using Git, GitHub, and VS Code.

Meteorologist & Secretary to the Deputy Director-General for Academic Affairs | Thai Meteorological Dept., Thailand
May 2006 – May 2018

- Developed predictive models and analyzed large-scale climate data to provide actionable insights.
- Coordinated inter-agency academic projects, advised executive leadership, and managed project lifecycles by reviewing, summarizing, and drafting technical reports and policy.